

Ecommerce Analytics Project Report

Project Overview

This project was conducted to evaluate the performance of an ecommerce business by analyzing transactional data and customer behavior. Using SQL-driven analysis, the study focused on identifying revenue drivers, customer retention patterns, and growth opportunities. The objective was to transform raw data into actionable insights that support strategic business decisions.

Dataset Description

The dataset includes four integrated tables: customers, orders, order details, and products. These tables were organized using a star schema model to enable efficient relational analysis. The dataset captures sales transactions, customer attributes, and product performance metrics, allowing comprehensive business evaluation.

Key Findings

The analysis revealed several critical insights:

- Total company revenue reached approximately ₹19.78 million.
- Average customer lifetime value was approximately ₹235,512.
- Nearly 69% of customers were repeat buyers, demonstrating strong retention.
- Revenue trends showed significant monthly volatility, with a major spike observed in July.
- A small segment of high-value customers generated a disproportionate share of total revenue.
- Chennai emerged as the highest revenue-generating city, followed by Delhi and Lucknow.
- Customers placed an average of 2.38 orders, reflecting healthy purchasing behavior.

Business Recommendations

Based on the findings, the following strategic actions are recommended:

- Develop targeted loyalty programs for high-value customers to maximize retention and profitability.
- Prioritize marketing investments in top-performing cities such as Chennai.
- Implement demand forecasting to manage revenue volatility.
- Strengthen promotion of top-selling premium products.
- Apply customer segmentation to personalize engagement strategies.

Tools and Technologies Used

This project utilized MySQL for SQL-based analytics, including advanced joins, aggregations, and window functions. Power BI was used to design an executive dashboard for visualizing business insights.

Conclusion

This project demonstrates the practical application of end-to-end ecommerce analytics in converting raw data into meaningful business intelligence. The insights generated highlight opportunities for revenue optimization, customer retention, and strategic expansion, reflecting real-world analytical practices used in modern organizations.